

UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : PHAS0042

ASSESSMENT PATTERN: PHAS0042A6UF 001

MODULE NAME : PHAS0042 - Quantum Mechanics

LEVEL : Undergraduate

DATE: : 07/05/2024

TIME : 14:30

DURATION : 3 Hours

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

**Year
2023/24**

**EXAMINATION PAPER CANNOT BE REMOVED FROM THE EXAM HALL. PLACE
EXAM PAPER AND ALL COMPLETED SCRIPTS INSIDE THE EXAMINATION
ENVELOPE**

Hall instructions	
Additional materials	
Standard Calculators	Yes
Non-Standard Calculators	

TURN OVER

Answer All Three Questions

The numbers in square brackets show the provisional allocation of maximum marks per question or part of question.

A module materials booklet, including physical constants if applicable, is attached at the end of the paper.

[Part marks]

1. (a) The sets of states $\{|\chi_m\rangle\}$ and $\{|\psi_n\rangle\}$ are two independent orthonormal sets of basis states such that a quantum state $|\Psi\rangle$ can be written $|\Psi\rangle = \sum_n c_n |\psi_n\rangle$ and $|\Psi\rangle = \sum_m d_m |\chi_m\rangle$. Use the closure relation to show that the change in representation between column vectors c and d can be expressed as $d = Sc$ where S is a matrix with elements $S_{mn} = \langle\chi_m|\psi_n\rangle$.

[5]

- (b) Show that the matrix representation A_χ of an operator $\hat{A}$ in the $\{|\chi_m\rangle\}$ basis is related to A_ψ , its representation in the $\{|\psi_n\rangle\}$ basis, by

[4]

$$A^{(\chi)} = SA^{(\psi)}S^\dagger.$$

- (c) A spin- $\frac{1}{2}$ particle has a Hamiltonian $\hat{H} = \gamma [a\hat{S}_x + b\hat{S}_y + c\hat{S}_z]$ with real coefficients a, b, c and $a^2 + b^2 + c^2 = 1$; γ is a positive constant and $\hat{S}_x, \hat{S}_y, \hat{S}_z$ are the operators for the Cartesian components of spin.

Labelling the eigenstates of $\hat{S}_z$ as $\{|\alpha\rangle, |\beta\rangle\}$ show that the matrix representing the Hamiltonian in the basis $\{|\alpha\rangle, |\beta\rangle\}$ is

[3]

$$H^{(z)} = \gamma \frac{\hbar}{2} \begin{pmatrix} c & a - ib \\ a + ib & -c \end{pmatrix}.$$

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[Part marks]

(Question continued from previous page)

- (d) The eigenstates of $\hat{S}_y$ are $|+_y\rangle = \frac{1}{\sqrt{2}}(|\alpha\rangle + i|\beta\rangle)$ and $|-_y\rangle = \frac{1}{\sqrt{2}}(|\alpha\rangle - i|\beta\rangle)$ with eigenvalues $\hbar/2$ and $-\hbar/2$ respectively. Show that the matrix $S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}$ can be used to transform a representation of a state in the $\{|\alpha\rangle, |\beta\rangle\}$ basis to a representation in the $\{|+_y\rangle, |-_y\rangle\}$ basis. [4]
- (e) Hence show that the matrix representing $\hat{H}$ in the $\{|+_y\rangle, |-_y\rangle\}$ basis is

$$H^{(y)} = \gamma \frac{\hbar}{2} \begin{pmatrix} b & c - ia \\ c + ia & -b \end{pmatrix}. \quad [4]$$

2. Consider a system of two spin-half particles, such that particle 1 has the spin quantum number $s_1 = \frac{1}{2}$ and particle 2 has $s_2 = \frac{1}{2}$. Let S and M be the quantum numbers of the total spin for the joint system.

For these questions, you may find it useful to refer to the Clebsch-Gordan Coefficients table in the materials booklet attached at the end of the paper.

- (a) For the state $|S = 0, M = 0\rangle$, calculate $\langle \hat{S}_{z1} \rangle$, where $\hat{S}_{z1}$ is the operator for the z-component of spin for particle 1. [3]
- (b) For these particles, show that:

$$\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 = \frac{1}{2} (\hat{S}^2 - \hat{S}_1^2 - \hat{S}_2^2),$$

where $\hat{\mathbf{S}} = \hat{\mathbf{S}}_1 + \hat{\mathbf{S}}_2$. [2]

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[Part marks]

(Question continued from previous page)

(c) The particles have the following Hamiltonian:

$$\hat{H} = \lambda \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 + B_z \hat{S}_{z1} + B_z \hat{S}_{z2}$$

where B_z is the magnitude of an external magnetic field and λ is a real parameter. Show that the matrix representation of this Hamiltonian with respect to the common eigenbasis of $\hat{S}_{z1}$ and $\hat{S}_{z2}$ is given by:

$$H = \begin{pmatrix} \frac{\lambda}{4}\hbar^2 + B_z\hbar & 0 & 0 & 0 \\ 0 & -\frac{\lambda}{4}\hbar^2 & \frac{\lambda}{2}\hbar^2 & 0 \\ 0 & \frac{\lambda}{2}\hbar^2 & -\frac{\lambda}{4}\hbar^2 & 0 \\ 0 & 0 & 0 & \frac{\lambda}{4}\hbar^2 - B_z\hbar \end{pmatrix}$$

Define all terms and work consistently in one basis set where appropriate.

[8]

(d) Find the eigenvalues and eigenvectors of the Hamiltonian above by reordering the basis such that it forms two 2-by-2 matrices.

[7]

CONTINUED

[Part marks]

3. A particle in a harmonic potential has Hamiltonian

$$\hat{H}_0 = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right)$$

The potential then is modified by a weak linear field, resulting in the Hamiltonian;

$$\hat{H} = \hat{H}_0 + \hat{W} = \hbar\omega \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) + \lambda \hat{x} .$$

- (a) Using perturbation theory, compute the first order corrections to the energy for all eigenstates of $\hat{H}_0$. [3]
- (b) Compute the second order correction to the energy of *the ground state only*. [3]
- (c) The ground state energy can also be approximated using the *variational method*. Considering a normalised trial state of the form:

$$|\psi_\theta\rangle = \cos(\theta)|0\rangle + \sin(\theta)|1\rangle,$$

where θ is a real parameter, show that the expectation value of energy for $|\psi_\theta\rangle$ is: [4]

$$\langle \hat{H} \rangle_\theta = \frac{\hbar\omega}{2} \left(\cos^2(\theta) + 3 \sin^2(\theta) \right) + \frac{\sqrt{2}\lambda}{\alpha} \cos(\theta) \sin(\theta).$$

- (d) Show that this expectation is minimised when the following equation holds:

$$\tan(2\theta) = -\frac{\sqrt{2}\lambda}{\alpha\hbar\omega} .$$

You may find the following identities helpful: [3]

$$\sin(2x) = 2 \sin(x) \cos(x) \quad \cos(2x) = \cos^2(x) - \sin^2(x)$$

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[Part marks]

(Question continued from previous page)

- (e) Assuming that $\lambda \ll (\alpha \hbar \omega)/(\sqrt{2})$, use the variational method to provide an upper bound to the ground state energy of $\hat{H}$ when $\hbar \omega = 1$ and $\lambda/\alpha = 0.01$. [6]
- (f) Compare the results of your variational method calculation with those from your perturbation theory calculation. Comment on any differences. [1]

END OF EXAMINATION PAPER

PHAS0042 Materials Booklet 2023-24

1 Formal Quantum Mechanics

We often represent state vectors in an orthonormal basis, $|\phi_n\rangle$, $n = 1, \dots, d$ where $\langle\phi_n|\phi_m\rangle = \delta_{n,m}$.

$$|\psi\rangle = \sum_n c_n |\phi_n\rangle.$$

Orthonormal basis sets satisfy the closure relation (also called the resolution of the identity):

$$\mathbb{I} = \sum_n |\phi_n\rangle\langle\phi_n|.$$

We can represent $|\psi\rangle$ with a column vector

$$|\psi\rangle \rightarrow \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \end{bmatrix}$$

Similarly, any bra vector can be represented as a row vector

$$\langle\psi| = \sum_n c_n^* \langle\phi_n| \quad \langle\psi| \rightarrow [c_1^* \quad c_2^* \quad c_3^* \quad \dots]$$

and any operator $\hat{A}$ can be represented by a matrix A , where the matrix elements of A are given by

$$A_{j,k} = \langle\phi_j|\hat{A}|\phi_k\rangle.$$

2 The Quantum Harmonic Oscillator

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi_n(x) = E_n \psi_n(x).$$

Its (orthonormal) eigenfunctions are:

$$\psi(x) = N_n(\alpha) H_n(\alpha x) e^{-\alpha^2 x^2/2}.$$

where $H_n(\alpha x)$ is Hermite polynomial of degree n , parity $(-1)^n$ while

$$N_n = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{1/2}, \text{ and } E_n = \left(n + \frac{1}{2} \right) \hbar\omega$$

$$\text{and } [\hat{x}, \hat{p}] = i\hbar$$

Raising and lowering operators

$$\hat{a}_+ = \frac{1}{\sqrt{2}} \left[\alpha \hat{x} - \frac{i}{\hbar\alpha} \hat{p} \right] \quad \text{and} \quad \hat{a}_- = \frac{1}{\sqrt{2}} \left[\alpha \hat{x} + \frac{i}{\hbar\alpha} \hat{p} \right], \text{ where } \alpha = (m\omega/\hbar)^{1/2}$$

$$\text{For arbitrary operators, } [\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}] \hat{C} + \hat{B} [\hat{A}, \hat{C}]$$

Key equations

$$\begin{aligned}\hat{a}_+^\dagger &= \hat{a}_-; & \hat{a}_-^\dagger &= \hat{a}_+; & [\hat{a}_-, \hat{a}_+] &= 1. \\ \hat{H} &= \left(\hat{a}_+ \hat{a}_- + \frac{1}{2} \right) \hbar \omega; & \hat{N} |n\rangle &= n |n\rangle \\ \hat{a}_- |n\rangle &= \sqrt{n} |n-1\rangle \\ \hat{a}_+ |n\rangle &= \sqrt{n+1} |n+1\rangle \\ (\Delta A)(\Delta B) &\geq \frac{1}{2} | \langle [\hat{A}, \hat{B}] \rangle | \end{aligned}$$

where:

$$(\Delta A)^2 = \langle \psi | (\hat{A} - \langle \hat{A} \rangle)^2 | \psi \rangle$$

3 & 4 Angular momentum and spin

The time-independent Schrödinger equation is

$$\hat{L}^2 Y_{lm}(\theta, \phi) = l(l+1) \hbar^2 Y_{lm}(\theta, \phi) \text{ and } \hat{L}_z Y_{lm}(\theta, \phi) = m \hbar Y_{lm}(\theta, \phi)$$

$$[\hat{L}_i, \hat{L}^2] = 0 \text{ for } i = x, y, z \quad [\hat{L}_x, \hat{L}_y] = i \hbar \hat{L}_z \text{ and cyclic permutations}$$

We convert cartesian coordinates to spherical polars:

$$x = r \sin \theta \cos \phi; \quad y = r \sin \theta \sin \phi \quad z = r \cos \theta$$

Angular momentum with raising and lowering operators

$$[\hat{J}_i, \hat{J}_j] = i \hbar J_k \text{ for cyclic permutations of } i, j, k \equiv x, y, z$$

$$\hat{J}_\pm = \hat{J}_x \pm i \hat{J}_y$$

Key equations

$$\begin{aligned}(\hat{J}_\pm)^\dagger &= \hat{J}_\mp; & \hat{J}_\pm \hat{J}_\mp &= \hat{J}^2 - \hat{J}_z^2 \pm \hbar \hat{J}_z \\ [\hat{J}_\pm, \hat{J}_\mp] &= \pm 2 \hbar \hat{J}_z; & [\hat{J}_z, \hat{J}_\pm] &= \pm \hbar \hat{J}_\pm \\ \hat{J}^2 |jm\rangle &= j(j+1) \hbar^2 |jm\rangle & \text{and } \hat{J}_z |jm\rangle &= m \hbar |jm\rangle\end{aligned}$$

$$\hat{J}_- |jm\rangle = \sqrt{j(j+1) - m(m-1)} \hbar |jm-1\rangle \quad (1)$$

$$\hat{J}_+ |jm\rangle = \sqrt{j(j+1) - m(m+1)} \hbar |jm+1\rangle \quad (2)$$

Spin-half operators can be written with respect to the $|\alpha\rangle, |\beta\rangle$ basis $\hat{S}_j = \frac{\hbar}{2} \hat{\sigma}_j$ where the Pauli matrices take the form:

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The eigenstates of the operator $\hat{n} \cdot \hat{\sigma}$ may be written:

$$|+\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} \sin \frac{\theta}{2} \\ -\cos \frac{\theta}{2} e^{i\phi} \end{pmatrix}$$

Adding Angular momenta

For a total angular momentum $\hat{\mathbf{J}} = \hat{\mathbf{J}}_1 + \hat{\mathbf{J}}_2$, we have eigenstates

$$\hat{\mathbf{J}}^2 |JM\rangle = J(J+1)\hbar^2 |JM\rangle \quad \text{and} \quad \hat{\mathbf{J}}_z |JM\rangle = M\hbar |JM\rangle$$

that are constructed using Clebsch Gordan coefficients $\langle j_1 m_1 j_2 M - m_1 | JM; j_1 j_2 \rangle$:

$$|JM\rangle \equiv |JM; j_1 j_2\rangle = \sum_{m_1} \langle j_1 m_1 j_2 M - m_1 | JM; j_1 j_2 \rangle |j_1 m_1\rangle |j_2 M - m_1\rangle,$$

A table of **Clebsch-Gordan coefficients** is on the last page of this document.

Source: https://en.wikipedia.org/wiki/Table_of_Clebsch-Gordan_coefficients

5 Quantum Mechanics with Many Particles

CHSH-Bell inequality

$$E(C) = E(a_0 b_0) + E(a_0 b_1) + E(a_1 b_0) - E(a_1 b_1) \leq 2$$

Slater Determinant

$$\Psi_{n_1 n_2 \dots n_N}^{(A)}(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_{n_1}(r_1) & \phi_{n_1}(r_2) & \cdots & \phi_{n_1}(r_N) \\ \phi_{n_2}(r_1) & \phi_{n_2}(r_2) & \cdots & \phi_{n_2}(r_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{n_N}(r_1) & \phi_{n_N}(r_2) & \cdots & \phi_{n_N}(r_N) \end{vmatrix}$$

6 Approximate Methods

Perturbation Theory - nondegenerate form

We have a Hamiltonian of the form:

$$\hat{H} = \hat{H}_0 + \lambda \hat{W}$$

where $\hat{H}_0$ is non-degenerate and $\hat{H}_0 |\psi_n^{(0)}\rangle = E_n^{(0)} |\psi_n^{(0)}\rangle$.

We form power series expansions:

$$E_n = \sum_{j=0}^{\infty} \lambda^j E_n^{(j)} = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \quad |\psi_n\rangle = \sum_{j=0}^{\infty} \lambda^j |\psi_n^{(j)}\rangle = |\psi_n^{(0)}\rangle + \lambda |\psi_n^{(1)}\rangle + \lambda^2 |\psi_n^{(2)}\rangle + \dots$$

The first order correction to the eigenvalues are:

$$E_n^{(1)} = \langle \psi_n^{(0)} | \hat{W} | \psi_n^{(0)} \rangle.$$

The first order correction to the eigenvectors is:

$$|\psi_n^{(1)}\rangle = \sum_{k \neq n} \frac{W_{kn}}{(E_n^{(0)} - E_k^{(0)})} |\psi_k^{(0)}\rangle.$$

where here $W_{\ell n} = \langle \psi_\ell^{(0)} | \hat{W} | \psi_n^{(0)} \rangle$.

The second order correction to the eigenvalues is:

$$E_n^{(2)} = \sum_{k \neq n} \frac{|W_{kn}|^2}{(E_n^{(0)} - E_k^{(0)})} = \sum_{k \neq n} \frac{|\langle \psi_k^{(0)} | \hat{W} | \psi_n^{(0)} \rangle|^2}{(E_n^{(0)} - E_k^{(0)})}.$$

Perturbation Theory - degenerate form

We have a Hamiltonian of the form:

$$\hat{H} = \hat{H}_0 + \lambda \hat{W}$$

where $\hat{H}_0$ is degenerate and we label its degenerate eigenstates $\hat{H}_0 |\chi_{n\mu}\rangle = E_n |\chi_{n\mu}\rangle$.

Again, we expand eigenvalues and eigenvectors as a power series

$$|\psi_{n\mu}\rangle = |\psi_{n\mu}^{(0)}\rangle + \lambda |\psi_{n\mu}^{(1)}\rangle + \lambda^2 |\psi_{n\mu}^{(2)}\rangle + \dots, \quad E_{n\mu} = E_n^{(0)} + \lambda E_{n\mu}^{(1)} + \lambda^2 E_{n\mu}^{(2)} + \dots$$

We compute the first order eigenvalue correction and identify the zeroth order eigenvector term by solving a matrix equation. For example, for a 2-fold degenerate eigenvalue of $\hat{H}_0$, it has the form:

$$\begin{pmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{pmatrix} \begin{pmatrix} c_{j1} \\ c_{j2} \end{pmatrix} = E_{nj}^{(1)} \begin{pmatrix} c_{j1} \\ c_{j2} \end{pmatrix}.$$

for $j = 1, 2$ where here $W_{ij} = \langle \chi_{ni} | \hat{W} | \chi_{nj} \rangle$ and where

$$|\psi_{nj}^{(0)}\rangle = c_{j1} |\chi_{n1}\rangle + c_{j2} |\chi_{n2}\rangle$$

Variational Method

The variational principle states that

$$\frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0.$$

where $|\psi\rangle$ is any state, which does not have to be normalised.

The variational method allows us to derive an upper bound of (and hence an estimate of) the ground state energy by minimising the energy of a parameterised trial solution $|\psi_{\vec{\alpha}}\rangle = |\psi_T(\alpha_1, \alpha_2, \dots, \alpha_s)\rangle$.

We thus compute

$$E(\alpha_1, \alpha_2, \dots, \alpha_s) = E(\vec{\alpha}) = \langle \hat{H} \rangle_{\vec{\alpha}} = \frac{\langle \psi_{\vec{\alpha}} | \hat{H} | \psi_{\vec{\alpha}} \rangle}{\langle \psi_{\vec{\alpha}} | \psi_{\vec{\alpha}} \rangle}.$$

and use calculus to determine the minimum value of $E(\vec{\alpha})$.

CLEBSCH-GORDAN COEFFICIENTS

$$j_1 = \frac{1}{2}, j_2 = \frac{1}{2} \quad [\text{edit}]$$

$$m = 1$$

m_1, m_2	j	1
$\frac{1}{2}, \frac{1}{2}$	1	1

$$m = -1$$

m_1, m_2	j	1
$-\frac{1}{2}, -\frac{1}{2}$	1	1

$$m = 0$$

m_1, m_2	j	1	0
$\frac{1}{2}, -\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
$-\frac{1}{2}, \frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$$j_1 = 1, j_2 = \frac{1}{2} \quad [\text{edit}]$$

$$m = \frac{3}{2}$$

m_1, m_2	j	$\frac{3}{2}$	$\frac{1}{2}$
$1, \frac{1}{2}$	1	1	1

$$m = \frac{1}{2}$$

m_1, m_2	j	$\frac{3}{2}$	$\frac{1}{2}$
$1, -\frac{1}{2}$	$\sqrt{\frac{1}{3}}$	$\sqrt{\frac{1}{3}}$	$\sqrt{\frac{2}{3}}$
$0, \frac{1}{2}$	$\sqrt{\frac{2}{3}}$	$\sqrt{\frac{2}{3}}$	$-\sqrt{\frac{1}{3}}$

For brevity, solutions with $M < 0$ and $j_1 < j_2$ are omitted. They may be calculated using the simple relations

$$\langle j_1, j_2; m_1, m_2 \mid j_1, j_2; J, M \rangle = (-1)^{J-j_1-j_2} \langle j_1, j_2; -m_1, -m_2 \mid j_1, j_2; J, -M \rangle.$$

and

$$\langle j_1, j_2; m_1, m_2 \mid j_1, j_2; J, M \rangle = (-1)^{J-j_1-j_2} \langle j_2, j_1; m_2, m_1 \mid j_2, j_1; J, M \rangle.$$

$$j_1 = \frac{3}{2}, j_2 = \frac{1}{2} \quad [\text{edit}]$$

$$m = 2$$

m_1, m_2	j	2
$\frac{3}{2}, \frac{1}{2}$	1	1

$$m = 1$$

m_1, m_2	j	2	1
$\frac{3}{2}, -\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\sqrt{\frac{3}{4}}$
$1, \frac{1}{2}$	$\sqrt{\frac{3}{4}}$	$\sqrt{\frac{3}{4}}$	$-\frac{1}{2}$

$$m = 0$$

m_1, m_2	j	2	1
$\frac{1}{2}, -\frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
$1, \frac{1}{2}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$$j_1 = 1, j_2 = 1 \quad [\text{edit}]$$

$$m = 2$$

m_1, m_2	j	2
1, 1	1	1

$$m = 1$$

m_1, m_2	j	2	1
1, 0	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$
0, 1	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{2}}$	$-\sqrt{\frac{1}{2}}$

$$m = 0$$

m_1, m_2	j	2	1	0
1, -1	$\sqrt{\frac{1}{6}}$	$\sqrt{\frac{1}{6}}$	$\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$
0, 0	$\sqrt{\frac{2}{3}}$	$\sqrt{\frac{2}{3}}$	0	$-\sqrt{\frac{1}{3}}$
-1, 1	$\sqrt{\frac{1}{6}}$	$\sqrt{\frac{1}{6}}$	$-\sqrt{\frac{1}{2}}$	$\sqrt{\frac{1}{3}}$